

SMART HOME LIGHTING SYSTEM

Description: A Smart Home Lighting System automatically controls lights based on motion detection and ambient light conditions. The system uses sensors and a microcontroller to improve energy efficiency and user convenience.

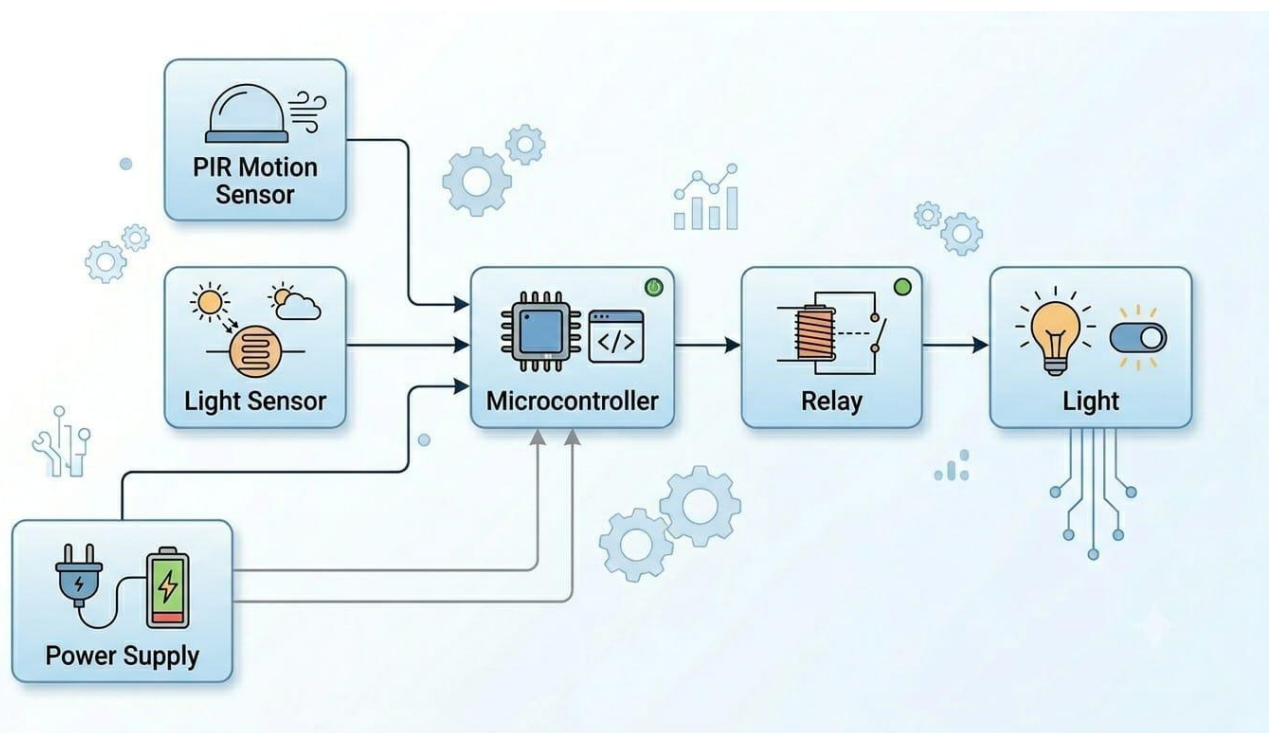
Working Principle:

- ✓ PIR sensor detects human movement.
- ✓ Light sensor checks the surrounding brightness.
- ✓ The microcontroller processes sensor data.
- ✓ If motion is detected in a dark environment, the light turns ON.
- ✓ When no motion is detected for a specific time, the light turns OFF automatically.

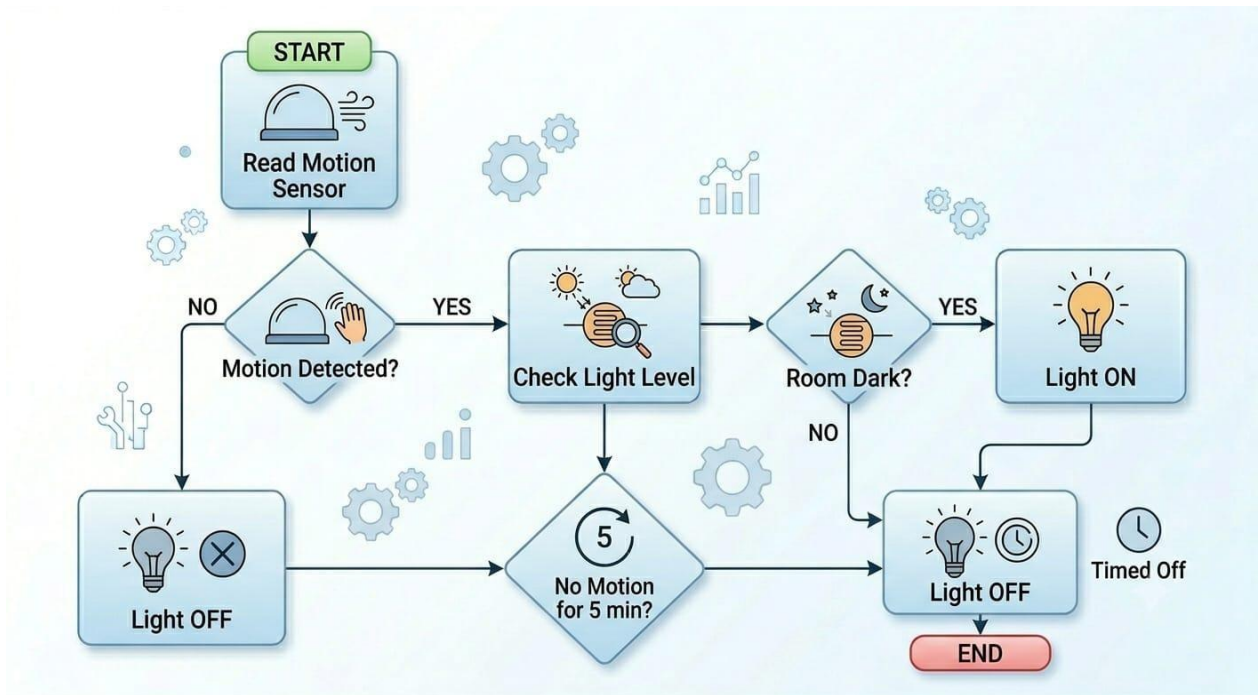
Components Used:

- ✓ PIR Motion Sensor
- ✓ LDR Light Sensor
- ✓ Microcontroller (Arduino/ESP32)
- ✓ Relay Module
- ✓ LED Light
- ✓ Power Supply

Block Design:



Flowchart:



Advantages:

- ✓ Saves electrical energy
- ✓ Reduces manual operation
- ✓ Improves home automation
- ✓ Enhances user comfort and safety